Seong-Jun Kang, Ph.D.

*Immunologist working on single-cell and multiplexed measurement of immune responses, in transplantation, neuroinflammation, and inflammatory disease*

Postdoctoral Fellow applicant · BioTranslation Lab, Center for Nanomedicine, Brigham and Women's Hospital and Harvard Medical School (RQ4074476, Position 3: Immunology and Immunoengineering)

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### PROFESSIONAL STATUS

I am an immunologist, and I measure what the immune system does after an intervention. During my doctoral training I worked in a laboratory that ran preclinical islet transplantation in non-human primates, and eight of my publications come from that work. At PB Immune Therapeutics I led an immune-regulatory antibody program, built the in-house single-cell and spatial analysis pipeline, and designed and ran EAE studies in cynomolgus monkeys and marmosets.

### RESEARCH OBJECTIVE

to measure what the immune system does after an intervention, one cell at a time, and to check that the result still holds on data the model has not seen. In transplantation that has meant asking how narrow the suppression can be and still keep the graft.

### SCIENTIFIC INTERESTS

immunomodulation and host response after an intervention; local versus systemic immunosuppression in transplantation; neuroinflammation and the blood-CSF barrier; high-parameter flow cytometry and cytokine profiling; single-cell and spatial analysis of tissue immune compartments; cohort and batch structure in in vivo study design

## WHAT I WOULD BRING

- **Immunosuppression for transplantation.** I am a co-author on non-human-primate islet transplantation studies of anti-CD40 (2C10R4) regimens, of a JAK inhibitor used in place of tacrolimus, and of IL-6 blockade. I am co-first author on long-term diabetes control with a tofacitinib-based regimen in rhesus monkeys, and a co-inventor on a humanized anti-CD40 antibody patent from 2024. That work was on the systemic side of immunosuppression, and materials and drug delivery would be new to me.
- **Neuroinflammation.** I did whole-brain immune cell isolation through to single-cell analysis myself: enzymatic digestion, Percoll gradient, FACS, 10×, and the analysis. From 11,587 cells in an Alzheimer's disease mouse model I resolved ten immune populations. I reported exhausted CD8+ tissue-resident memory T cells interacting with the choroid plexus epithelium, the blood-CSF barrier.
- **High-parameter flow cytometry and cytokine profiling.** I run high-dimensional flow cytometry with panels of up to 24 markers, and I design the panels myself. I have also run 10-color six-way sorting. For plasma proteins and cytokines I use Olink targeted proteomics with the immune and neurology panels.
- **Single-cell and spatial analysis at scale.** I work with scRNA-seq, scTCR-seq, snATAC-seq, GeoMx spatial transcriptomics and Visium, at the bench and in code. I decomposed cell subsets at 251,055-cell scale with consensus NMF, and integrated cohorts across more than 700 donors, 8 organs and 23 immune diseases. I check the result with held-out AUC and calibration.
- **Studies collected in batches.** I put reference material in every collection round and write batch into the model as a variable. In nested designs I test at the level of the donor or the animal rather than the cell.

## EDUCATION

DEGREES

**Ph.D.** **Seoul National University**, Seoul, Republic of Korea
2014–2023 Department of Biomedical Sciences
*Thesis: Exploration of the Role of Regulatory T cells in Various Pathogenic Contexts*
Advisor: Chung-Gyu Park, M.D., Ph.D.

**M.S.** **2012–2014**
**Seoul National University**, Seoul, Republic of Korea
Department of Biomedical Sciences
*Thesis: Enhanced LPS-Induced Monocyte Activation by Recombinant Human Soluble CD14*
Advisor: Chung-Gyu Park, M.D., Ph.D.

**B.S.** **2008–2012**
**Sungkyunkwan University**, Suwon, Republic of Korea
Department of Genetic Engineering

## RESEARCH & PROFESSIONAL EXPERIENCE

**Researcher (postdoctoral)** **2025–present**
**Yonsei University Wonju College of Medicine** · Dept. of Physiology / Organelle Medicine Research Center, Wonju, Republic of Korea
Deep-learning multimodal-integration models fusing imaging, transcriptomic, and clinical data into single predictive models.

**Associate Director** **2023–2025**
**PB Immune Therapeutics Inc.** · Dept. of Basic Research, Seoul, Republic of Korea
Led a humanized anti-CD40 antibody program: candidate ranking by **surface plasmon resonance (Biacore)** and ELISA, receptor-blocking confirmation, human and non-human-primate cross-reactivity, in vivo efficacy in primates, and the regulatory documentation. Placed and oversaw CRO and core-facility work, from study design and assay conditions through data delivery and go/no-go calls. Built the in-house single-cell and spatial analysis pipeline. Ran cynomolgus and marmoset EAE studies for multiple sclerosis programs.

**Manager** **2021–2023**
**PB Immune Therapeutics Inc.** · Dept. of Basic Research, Seoul, Republic of Korea
Wrote the two KDDF proposals that funded the anti-CD40 program and ran the lead-selection work under them.

**Research Student** **2020–2023**
**SCAID, Single Cell Atlas of Immune Diseases**, Genomic Medicine Institute, Seoul National University, Seoul, Republic of Korea
Contributed the founding proposal; executed within the funded program (700+ donors × 8 organs): milestone tracking, multi-site QC, multi-PI coordination.

**Researcher** **2019–2021**
**Xenotransplantation Research Center, Seoul National University**, Seoul, Republic of Korea
Rhesus macaque tissue processing and longitudinal immune monitoring across transplantation studies.

**Teaching Assistant** **2014–2016**
**Seoul National University College of Medicine** · Dept. of Microbiology and Immunology, Seoul, Republic of Korea

## TECHNICAL SKILLS

**Transplantation & in vivo immunology**
Non-human-primate islet transplantation, systemic immunosuppression side: co-author on studies of **anti-CD40 (2C10R4)**, of a **JAK inhibitor used in place of tacrolimus**, and of **IL-6 blockade**; co-first author on long-term diabetes control with a tofacitinib-based regimen in rhesus monkeys (*Xenotransplantation* 2024). Tissue processing, longitudinal immune monitoring, and brain and spinal cord harvest on those studies. **EAE** studies in cynomolgus monkeys and marmosets designed and run for multiple sclerosis programs. Mouse **Alzheimer's disease** neuroimmune profiling: whole-brain digestion, Percoll gradient, $CD45^{+}$ FACS, 10×, analysis (*J Neuroimmunol* 2025, co-first author).

**Binding & protein interaction**
**Surface plasmon resonance (Biacore)** run hands-on for antibody lead work: association and dissociation rates, affinity ranking across a candidate panel, receptor-blocking confirmation, and human / non-human-primate cross-reactivity. **ELISA**-based affinity and epitope analysis; competition binding. Antibody lead selection and optimization from candidate panel to registered patent. Functional validation of a peptide inhibitor of calcineurin in primary human T cells (CABIN1 core peptide, *Exp Mol Med* 2022).

| | |
|---|---|
| **Program & external execution** | CRO and core-facility work from study design and placement through data delivery and go/no-go calls; assay conditions specified to the vendor; regulatory documentation for a therapeutic antibody program; funder milestones and reporting; multi-site sample QC across a 700-donor collection. |
| **Immunological assays** | High-dimensional flow cytometry (panels up to 24 markers); multiplex panel design; 10-color 6-way FACS. **Olink** targeted plasma proteomics (immune & neurology panels). Primary human T-cell isolation, expansion and functional assays; monocyte and dendritic-cell differentiation; mesenchymal stem cell culture. Enzymatic tissue digestion, Percoll enrichment, cell and nucleus isolation. |
| **Single-cell / spatial / multi-omic** | scRNA-seq, scTCR-seq, snRNA/snATAC; 10× 5'/3' library generation and **10× Chromium Connect** automated preparation. **Seurat, Scanpy, Squidpy, Monocle3, CellChat/NicheNet**; scVI/scANVI, batch integration, **consensus NMF**; Milo KNN-graph differential abundance, GLIPH, InferCNV. Custom UMI-aware, BAM-level isoform analysis code (Python). Spatial: **GeoMx DSP** run end to end, **Visium**, **MACSima** cyclic multiplexed immunofluorescence. |
| **Computational** | Python: scikit-learn, XGBoost/LightGBM/CatBoost, **PyTorch**; AUC-validated classifiers held out on independent cohorts; multimodal deep-learning integration (imaging + transcriptomic + clinical). Linux servers · R/RStudio · Git-based reproducible analysis. |
| **Microscopy & animal models** | Spinning-disk confocal (Nikon CSU-W1 SORA), point-scanning confocal (Leica SP8), two-photon; IHC/IF (frozen & FFPE) with quantitative image analysis. Non-human primates across three species (rhesus, cynomolgus, marmoset), including EAE and islet-transplantation studies; mouse immune profiling. |

## PUBLICATIONS

| 18 | 17 | 1 | 6 |
|---|---|---|---|
| TOTAL PAPERS | PEER-REVIEWED | IN REVISION | FIRST / CO-FIRST |

**Google Scholar · Full Profile, Citations & h-index ↗**

SELECTED PUBLICATIONS · FIRST / CO-FIRST AUTHOR

**2026** Woogil Song*, **Seong-Jun Kang***, Taehee Stella Kim*, Seyoung Jung, Sung Hee Kim, Su Chan Park, Sung Hae Chang, Eun Young Lee, Tae-Gyun Kim, Hyun Je Kim, Jeong Seok Lee. **"Conserved PDE4D short-isoform bias defines pathogenic Th1.17 state across human inflammatory diseases."** ***Nature Immunology***. IF 27.7 in revision

*First author (co-first).* ***Design*** *— four matched cohorts (healthy control, psoriasis, rheumatoid arthritis, ankylosing spondylitis), 16 donors each, n = 64; memory $CD4^+$ T enrichment by negative selection, 4-plex pooling with SNP demultiplexing so donor and sequencing run are not confounded; 5′ scRNA-seq with paired TCR. 251,055 memory $CD4^+$ T cells.* ***Analysis*** *— built custom UMI-aware, BAM-level code (Python) resolving PDE4D short- vs long-isoform identity from 5′ reads, which standard count matrices cannot separate; consensus NMF (k = 14 → 12 programs); scRNA + scTCR + snATAC + STAT3 ChIP-seq integration; Milo KNN-graph differential abundance; condition classifier on Th1.17 subset composition — arthritis AUC 0.780 (95% CI 0.665–0.896), healthy control 0.771, psoriasis 0.806.* ***Generalisation*** *— the PDE4D-high Th1.17 state was conserved across the SCAID atlas (702 samples, 23 immune diseases) and in independent SLE and arthritis synovial-fluid datasets.* ***Causal test*** *— short-isoform overexpression (PDE4D1/2/6) raised the IFN-$\gamma^+$IL-$17A^+$ fraction while the long isoform PDE4D5 did not; the PDE4D-selective inhibitor zatolmilast lowered it.*

**2025** **Seong-Jun Kang***, Yong-Hee Kim*, Thuy Nguyen-Phuong*, Yijoon Kim, Jin-Mi Oh, Jae-chun Go, DaeSik Kim, Chung-Gyu Park, Hyunsu Lee, Hyun Je Kim. **"Immune cell-enriched single-cell RNA sequencing unveils the interplay between infiltrated CD8⁺ T resident memory cells and choroid plexus epithelial cells in Alzheimer's disease."** ***Journal of Neuroimmunology***. IF 2.9 Download PDF

*First author (co-first). Built the complete brain-immune single-cell analysis path (whole-brain harvest → enzymatic digestion → Percoll enrichment → FACS → 10x → analysis); identified exhausted CD8⁺ tissue-resident memory T cells engaging choroid-plexus epithelium (↑ MHC-I and interferon-stimulated genes) at the blood–CSF barrier. 11,587 cells, 10 immune populations.*

**2025** **Seong-Jun Kang***, Jeong-Ryeol Gong*, Seon-Pil Jin, Jin-Mi Oh, Hyunjin Jin, Yuji Lee, Yewon Moon, Dongjun Kim, Hyo Jeong Nam, Hyun Seung Choi, Sanha Hwang, Yun Jung Huh, Kyung Yeon Han, Jihwan Moon, Jongsuk Chung, Woong-Yang Park, Chung-Gyu Park, Hyun Je Kim, Jeong Eun Kim. **"Deciphering Dysfunctional Regulatory T Cells in Atopic Dermatitis."** ***Allergy***. IF 12.6 Download PDF

*First author (co-first). Single-cell and functional dissection of dysfunctional skin-homing (CLA⁺) regulatory T cells in atopic dermatitis; OX40 (TNFRSF4) up-regulated alongside impaired suppressive function.*

**2024** Jong-Min Kim*, **Seong-Jun Kang***, So-Hee Hong*, Hyunwoo Chung, Jun-Seop Shin, Byoung-Hoon Min, Hyun Je Kim, Jongwon Ha, Chung-Gyu Park. **"Long-term control of diabetes by tofacitinib-based immunosuppressive regimen after allo-islet transplantation in diabetic rhesus monkeys that rejected previously transplanted porcine islets."** ***Xenotransplantation***. IF 3.9 Download PDF

*First author (co-first). Long-term diabetes control with a tofacitinib-based regimen after allo-islet transplantation in rhesus macaques; longitudinal immune monitoring.*

**2022** Youngkyoung Lim*, **Seong-Jun Kang***, Beom Keun Cho, Hyun Je Kim, … Chong Hyun Won, Chung-Gyu Park. **"Intra-tumoral heterogeneity and immune escape of melanoma arising from congenital melanocytic nevus revealed by spatial gene expression profiling."** ***JEADV***. IF 9.2 Download PDF

*First author (co-first). Ran the GeoMx DSP workflow end to end — assay setup and instrument operation, immunofluorescence staining (S100B / CD3ε / nuclei), fluorescence-guided ROI selection and imaging, barcode collection, then the image and spatial-transcriptomic analysis.* ***Design*** *— one stage-IIC melanoma arising from a 4 cm congenital melanocytic nevus, segmented into four regions by invasion depth (M1–M4, 300 → 4,200 µm Breslow), four whole-transcriptome ROIs per region, 16 ROIs total, 18,677 genes. 2,067 differentially expressed genes (11.07%) at $P < 0.0005$; graded loss of HLA class I/II with depth, and B2M loss confirmed by immunohistochemistry.*

**2019** Jong-Min Kim*, Jun-Seop Shin*, Byoung-Hoon Min*, **Seong-Jun Kang***, Il-Hee Yoon, Hyunwoo Chung, Jiyeon Kim, Eung-Soo Hwang, Jongwon Ha, Chung-Gyu Park. **"JAK3 inhibitor-based immunosuppression in allogeneic islet transplantation in cynomolgus monkeys."** ***Islets***. IF 2.2 Download PDF

*Co-first author; one of four authors marked as contributing equally. JAK3-inhibitor-based immunosuppression in allogeneic islet transplantation in cynomolgus macaques.*

### CONTRIBUTING AUTHOR

**2026** Jung Ho Lee*, Sung Ha Lim*, Hyungdon Kook, **Seong-Jun Kang**, Geon Lee, Brian Hyohyoung Lee, Hyunsung Nam, YongJun Kim, Christine Suh-Yun Joh, Soyoung Jeong, Dongryeol Shin, Hyun Je Kim, Jiyoung Ahn. **"Mycosis Fungoides-like Atopic Dermatitis Represents a Th22-dominant Inflammatory Endotype."** ***Allergy***. IF 12.6

*Performed the scRNA-seq & scTCR-seq analysis (scVI cross-dataset integration with public MF/AD cohorts, InferCNV genomic-identity inference, scTCR clonality). Showed mfAD is a reactive, oligoclonal Th22 endotype rather than a malignancy, responsive to JAK1 inhibition. 248,881 cells.*

**2024** Brian Hyohyoung Lee*, Yoon Ji Bang*, Sung Ha Lim*, **Seong-Jun Kang**, Sung Hee Kim, Seunghee Kim-Schulze, Chung-Gyu Park, Hyun Je Kim, Tae-Gyun Kim. **"High-dimensional profiling of regulatory T cells in psoriasis reveals an impaired skin-trafficking property."** ***eBioMedicine***. IF 11.1 Download PDF

*Co-author. High-dimensional immune profiling of regulatory T cells in psoriasis, revealing an impaired skin-trafficking phenotype.*

2024 Youngkyoung Lim*, Beom Keun Cho*, **Seong-Jun Kang**, Soyoung Jeong, Hyun Je Kim, Jiyoon Baek, Ji Hwan Moon, Cheol Lee, Chan-Sik Park, Je-Ho Mun, Chong Hyun Won, Chung-Gyu Park. "**Spatial transcriptomic analysis of tumour–immune cell interactions in melanoma arising from congenital melanocytic nevus.**" ***JEADV***. IF 9.2 Download PDF

*Co-author. Spatial transcriptomic (GeoMx) analysis of tumour–immune cell interactions in nevus-arising melanoma.*

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2023 Sunyoung Jung*, Sunho Lee, Hyun Je Kim, Sueon Kim, Ji Hwan Moon, Hyunwoo Chung, **Seong-Jun Kang**, Chung-Gyu Park. "**Mesenchymal stem cell-derived extracellular vesicles subvert Th17 cells by destabilizing RORγt through posttranslational modification.**" ***Experimental & Molecular Medicine***. IF 12.8

*Co-author. MSC-derived extracellular vesicles suppress Th17 cells by destabilizing RORγt via post-translational modification.*

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2022 Sangho Lee, Han-Teo Lee, Young Ah Kim, Il-Hwan Lee, **Seong-Jun Kang**, Kyeongpyo Sim, Chung-Gyu Park, Kyungho Choi, Hong-Duk Youn. "**The optimized core peptide derived from CABIN1 efficiently inhibits calcineurin-mediated T-cell activation.**" ***Experimental & Molecular Medicine***. IF 12.8

*Co-author. A CABIN1-derived core peptide blocks the calcineurin–CABIN1 interaction and inhibits calcineurin-mediated T-cell activation. My part was the T-cell functional validation.*

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2021 So Hee Hong, Hyun Je Kim, **Seong-Jun Kang**, Chung-Gyu Park. "**Novel Immunomodulatory Approaches for Porcine Islet Xenotransplantation.**" ***Current Diabetes Reports***. IF 4.2 Download PDF

*Co-author. Review of novel immunomodulatory strategies for porcine islet xenotransplantation.*

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2020 Sunho Lee, Sueon Kim, Hyunwoo Chung, Ji Hwan Moon, **Seong-Jun Kang**, Chung-Gyu Park. "**Mesenchymal stem cell-derived exosomes suppress proliferation of T cells by inducing cell cycle arrest through p27kip1/Cdk2 signaling.**" ***Immunology Letters***. IF 4.4

*Co-author. MSC-derived exosomes suppress T-cell proliferation through p27kip1/Cdk2-mediated cell-cycle arrest.*

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2019 Hyun Je Kim, Ji Hwan Moon, Hyunwoo Chung, Jun-Seop Shin, Bongi Kim, Jong Min Kim, Jung-Sik Kim, Il-Hee Yoon, Byoung-Hoon Min, **Seong-Jun Kang**, Yong-Hee Kim, Kyuri Jo, Joungmin Choi, Heejoon Chae, Won Woo Lee, Sun Kim, Chung-Gyu Park. "**Bioinformatic analysis of peripheral blood RNA-sequencing sensitively detects the cause of late graft loss following overt hyperglycemia in pig-to-non-human primate islet xenotransplantation.**" ***Scientific Reports***. IF 4.6

*Co-author. Bioinformatic analysis of peripheral-blood RNA-seq to detect causes of late graft loss in pig-to-NHP islet xenotransplantation.*

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2018 Byoung-Hoon Min, Jun-Seop Shin, Jong-Min Kim, **Seong-Jun Kang**, Hyun-Je Kim, Il-Hee Yoon, Su-Kyoung Park, Ji-won Choi, Min-Suk Lee, Chung-Gyu Park. "**Delayed revascularization of islets after transplantation by IL-6 blockade in pig-to-non-human primate islet xenotransplantation model.**" ***Xenotransplantation***. IF 3.9

*Co-author. IL-6 blockade delays islet revascularization in a pig-to-NHP islet xenotransplantation model.*

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2018 Jun-Seop Shin, Jong-Min Kim, Byoung-Hoon Min, Il Hee Yoon, Hyun Je Kim, Jung-Sik Kim, Yong-Hee Kim, **Seong-Jun Kang**, Jiyeon Kim, Hee-Jung Kang, Dong-Gyun Lim, Eung-Soo Hwang, Jongwon Ha, Sang-Joon Kim, Wan Beom Park, Chung-Gyu Park. "**Pre-clinical results in pig-to-non-human primate islet xenotransplantation using anti-CD40 antibody (2C10R4)-based immunosuppression.**" ***Xenotransplantation***. IF 3.9

*Co-author. Pre-clinical pig-to-NHP islet xenotransplantation with anti-CD40 (2C10R4)-based immunosuppression.*

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2018 Jong-Min Kim, Jun-Seop Shin, Byoung-Hoon Min, Il Hee Yoon, **Seong-Jun Kang**, Won-Young Jeong, Sang-Joon Kim, Chung-Gyu Park. "**Pre-Clinical Results of Islet Allo-Transplantation Using JAK Inhibitor as Replacement for Tacrolimus in Cynomolgus Monkeys.**" ***Transplantation***. IF 6.2

*Co-author. Pre-clinical islet allo-transplantation using a JAK inhibitor in place of tacrolimus in cynomolgus macaques.*

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2018 Jung-Sik Kim, Hyunwoo Chung, Nari Byun, **Seong-Jun Kang**, Sunho Lee, Jun-Seop Shin, Chung-Gyu Park. "**Construction of EMSC-islet co-localizing composites for xenogeneic porcine islet transplantation.**" ***Biochemical and Biophysical Research Communications***. IF 3.1

*Co-author. Engineered EMSC-islet co-localizing composites for xenogeneic porcine islet transplantation.*

### MANUSCRIPTS IN PREPARATION

**2026** Kangsan Roh*, Sabyasachi Das*, **Seong-Jun Kang**, Olga Barreiro, Rodrigo Javier Gonzalez, Timothy P. Padera, Shraddha Shirguppe, Ahlim Lee, Youngju Song, Pazhanichamy Kalailingam, Claire E. Fong, Aarushi Gandhi, Meehyun Kim, Diana Tambala, Adam Vernon Crain, Casey A. Maguire, Ulrich H. Von Andrian, Christian L. Lino Cardenas, Sekeun Kim, Benjamin P. Kleinstiver, Mark E. Lindsay, Patricia L. Musolino, Rajeev Malhotra. **"In-vivo base editing of ACTA2 R179H restores lymphatic function and immune cell homeostasis in a murine model of Multisystemic Smooth Muscle Dysfunction Syndrome (MSMDS)."** in preparation

*Co-author (immunology). Performed the plasma proteomics analysis (Olink immune & neurology panels, 12 patients vs 6 controls), showing broad down-regulation of lymphoid regulators (LTβR, BAFF) and leukocyte-adhesion molecules, with dimensionality reduction separating patients from controls; complemented by flow-cytometry immunophenotyping of lymph-node leukocytes (germinal-center B cells GL7⁺B220⁺).*

## RESEARCH GRANTS & FUNDING

### INDUSTRY · PB IMMUNE THERAPEUTICS (MANAGER → ASSOCIATE DIRECTOR)

| FUNDING AGENCY / SPONSOR | PROJECT | PERIOD | BUDGET (USD) | ROLE / CONTRIBUTION |
|---|---|---|---|---|
| **Korea Drug Development Fund (KDDF)** | Lead selection and optimization of a therapeutic anti-CD40 antibody for the treatment of multiple sclerosis | 2021–2023 | **~$571,000** | *Lead proposal writer (>90% of proposal & execution)* |
| **Korea Drug Development Fund (KDDF)** | Evaluation of drug efficacy and discovery of pharmacodynamic biomarkers of a new anti-CD40 antibody lead compound (PB101) for the treatment of pemphigus vulgaris | 2022–2024 | **~$321,000** | *Lead proposal writer (>90% of proposal & execution)* |
| **Ministry of SMEs and Startups (MSS)** | Scale-up Funding for Tech Commercialization | 2022–2023 | **~$222,000** | *Lead proposal writer; managed >90% of execution* |
| **Ministry of SMEs and Startups (MSS)** | Early-stage startup R&D: anti-CD40 biologic (PB-40N) to prevent anti-drug antibodies | 2021–2022 | **~$74,000** | *Lead proposal writer & core researcher* |
| **NAVER Corporation** | Fundamental therapeutic strategy for immune-mediated diseases using engineered Tregs (PB20X_Treg) | 2023–2026 | **~$128,000** | *Proposal & study design (~50%)* |

### DOCTORAL TRAINING · SEOUL NATIONAL UNIVERSITY

| FUNDING AGENCY / SPONSOR | PROJECT | PERIOD | BUDGET (USD) | ROLE / CONTRIBUTION |
|---|---|---|---|---|
| **Ministry of Science and ICT (MSIT) · Bio & Medical Technology Development Program** | Establishing a meta-platform for discovering molecular targets by building a multi-layered immune-disease single-cell map (A9-22-1-05; SCAID, Single-cell Atlas for Immunological Disease) | 2022–2028 (3+3) | **~$1.6M** | *Participated in proposal writing (>30%)* |
| **Ministry of Education (MOE) & Ministry of Science and ICT (MSIT)** | Laboratory-Specialized Tech-Transfer & R&BD Grant | 2020–2021 | **~$40,000** | *Grant writing, experimental design & technical validation* |

*All grants were held by the institutional Principal Investigator (Ph.D. advisor at SNU; company CEO/CSO at PB Immune Therapeutics); the listed roles reflect the applicant's contribution to proposal writing and project execution,* ***not as PI****. Budgets converted at ≈ ₩1,400 / USD.*

## RESEARCH INTERESTS

- **Immune response after an intervention.** What the immune compartment does once a treatment goes in, measured cell by cell and over time. I have made that measurement in transplantation and in an Alzheimer's disease model.
- **Local versus systemic immunosuppression.** Whether suppression aimed at the graft leaves the rest of the immune system intact, with the local and the systemic compartment measured in the same animals. I have worked on the systemic side of that question in rhesus and cynomolgus monkeys.

- **Neuroinflammation and the blood-CSF barrier.** Which immune cells enter the brain, what state they arrive in, and what changes that state. In the Alzheimer's model I found exhausted CD8+ tissue-resident memory T cells interacting with the choroid plexus epithelium, with raised MHC class I and interferon-stimulated genes.
- **Multi-modal readouts of cell state.** Transcriptome, chromatin, spatial and plasma protein data brought into one cell-state model. I write the analysis code as well, including UMI-aware isoform analysis from 5' single-cell data.
- **Cohort structure in in vivo studies.** Animal studies arrive in cohorts, and cohort, age, surgery date and reagent lot all leave marks in the data. I have handled that structure across more than 700 donors, and the same arithmetic applies when a study has six animals per arm.

## AWARDS, PATENTS & SCHOLARSHIPS

### PATENTS

| | |
|---|---|
| 2024 | **Antagonistic Anti-CD40 Humanized Antibody.** Republic of Korea, Appl. No. 10-2024-0085972; registered 2024-07-01. Co-inventor. |
| 2019 | **Immunosuppression Composition Comprising JAK Inhibitor.** Republic of Korea, Appl. No. 10-2014-3490000; registered 2019-08-20. |

### AWARDS

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|---|---|
| 2024 | **Grand Prize**, 2024 GPTers AI Hackathon (Project GAIA Team) |
| 2023 | **Grand Prize**, SBA Smart Workathon (Work Process Innovation) |
| 2017 | **Best Poster Award**, American Transplant Congress, Chicago · **Best Presentation Award**, Korean Association of Immunologists |

### SCHOLARSHIPS

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|---|---|
| 2014–2020 | NRF Research Scholarship for Ph.D. Candidates · Dean's Lecture & Research Support Scholarship, SNU College of Medicine · BK21 Plus Research Scholarship |

## SELECTED PRESENTATIONS

**Poster** KAI Annual Meeting, Incheon (2022) · Cytokine, Hawaii (2022) · American Transplant Congress, Chicago (2017) · The Transplantation Society, Hong Kong (2016) · IPITA/IXA/CTS, Melbourne (2015) · TREG, Shanghai (2012)

## REFERENCES

**Chung-Gyu Park, M.D., Ph.D.**
Professor, Microbiology & Immunology / Biomedical Sciences, SNU College of Medicine; CEO/CSO, PB Immune Therapeutics · chgpark@snu.ac.kr
*Doctoral advisor and industry supervisor; directed the anti-CD40 antibody program.*

**Seung-Kuy Cha, Ph.D.**
Professor, Dept. of Physiology & Global Medical Science, Yonsei University Wonju College of Medicine · skcha@yonsei.ac.kr
*Most recent research supervisor (2025–2026).*

**Hyun Je Kim, M.D., Ph.D.**
Assistant Professor, Microbiology & Immunology / Biomedical Sciences; Interdisciplinary Program in Artificial Intelligence (IPAI), SNU · tte9801@snu.ac.kr
*Co-corresponding on the single-cell studies; supervised the computational work.*

**Jeong Seok Lee, M.D., Ph.D.**
Associate Professor, Graduate School of Medical Science & Engineering, KAIST, Daejeon · chemami@kaist.ac.kr
*Corresponding author, PDE4D / Nature Immunology study.*